

SPCE 5065
FINAL
SUMMER 2026

I completed this assignment using only authorized references. These are the textbook,
your personal notes, any material on the course Canvas page, and a calculator.

Signed:



Printed:

Jordan Clayton

1. ____ /10

2. ____ /10

3. ____ /10

4. ____ /8

5. ____ /8

6. ____ /10

7. ____ /9

8. ____ /10

9. ____ /15

10. ____ /10

Bonus: ____ /5

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1. (10 pts.) True/False

- a. True A spacecraft coated with a thermal material with a high emissivity will have a lower equilibrium temperature than material with a low emissivity.
- b. True Safe mode is often used to recover a spacecraft from a Micrometeoroid or Orbital Debris (MMOD) impact.
- c. False Free fall and zero gravity are the same.
- d. False For a spacecraft in GEO it is best to use a large positive bias voltage.
- e. False REMs and RADs are equivalent in the Van Allen Radiation belts.
- f. False The neutral environment, as explained in this course, is the same as the vacuum environment.
- g. True The most common contamination cleanliness level is class 10,000, which is equivalent to ISO level 7.
- h. True Galactic cosmic rays generally consist of higher-energy particles than those produced during solar particle events.
- i. False A Single Event Upset (SEU) permanently damages a microelectronic device and typically requires replacing the component.
- j. False A 1-mm debris particle in LEO is generally considered too small to cause significant spacecraft damage.

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2. (10 pts)

I. What is the relevance of a Single-Event Latch-up to a spacecraft's performance during a space mission?

- ☒ a. A latch-up can significantly impact spacecraft performance by causing excessive current draw, requiring protective power shutdown or reset, and potentially leading to permanent damage if not mitigated.
- b. A latch-up only temporarily impacts system performance because spacecraft fault-tolerance architectures always detect the event, isolate the affected circuit, and restore normal operation without risk of permanent damage.
- c. A latch-up catastrophically impacts system performance by immediately destroying the affected electronics, making the spacecraft or subsystem permanently inoperable without any possibility of recovery.
- d. A latch-up primarily impacts system performance by increasing the susceptibility of spacecraft electronics to subsequent single-event effects, resulting in progressively higher upset rates throughout the mission.
- e. None of the above.

II. Outgassing is commonly due to which of the following (*select all that apply*)

- a. decantation
- ☒ b. decomposition
- c. deflagration
- ☒ d. diffusion
- ☒ e. desorption

III. Solar ultraviolet radiation alters which of the following, which results in temperature changes on the spacecraft? (choose the *best* answer)?

- a. strength to flexibility ratio
- b. reflectivity to transmissivity ratio
- ☒ c. absorptivity to emissivity ratio
- d. stress to strain ratio
- e. decomposition to desorption ratio

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IV. The primary purpose of the thin front bumper in a Whipple shield is to

- a. Reflect the particle back into space.
- ☒ b. Vaporize or fragment the projectile before it reaches the rear wall.
- c. Absorb all of the impact energy by itself.
- d. Reduce spacecraft charging.
- e. Prevent thermal expansion.

V. The Half-Value Layer (HVL) of a material is defined as the thickness required to

- a. Reduce the radiation energy by one-half.
- ☒ b. Reduce the photon flux by one-half.
- d. Double the shielding effectiveness.
- d. Reduce the absorbed dose to zero.
- e. Eliminate secondary radiation.

3. (10 pts.) You are designing a *GEO satellite* with a sensor that will identify a target on earth's surface at 215 THz. Will this be an effective design? Explain your rationale. Hint: Consider both plasma oscillations (Lessons 7 & 8) and atmospheric transmittance (Lesson 2).

• wavelength: $\lambda = \frac{c}{f} = \frac{2.998 \times 10^8}{2.15 \times 10^{14}} = 1.394 \mu\text{m}$

• plasma check: $f_p = \frac{1}{2\pi} \sqrt{n_e e^2 / \epsilon_0 m_e} = 8.98 \sqrt{n_e} \text{ Hz}$

- $n_{e \text{ max}} = 4.5 \times 10^{12} \text{ m}^{-3}$

- $f_{p \text{ max}} = 8.98 \sqrt{4.5 \times 10^{12}} = 19.1 \text{ MHz} \ll 215 \text{ THz}$

The sensor is 10^7 times higher so plasma is irrelevant, however from the transmittance chart from neutral environment notes, there is a deep H₂O band that drops visibility to zero

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4. (8 pts.) Correlated the charged particle radiation flux density with solar activity by labelling “lower” or “higher.”

	Solar Min	Solar Max
Trapped radiation	-----	-----
- electrons	lower	higher
- protons	higher	lower
Galactic Cosmic Rays	higher	lower
Solar Particle Events	lower	higher

5. (8 pts) NO (Nitrous oxide) is a common element near earth. You may assume the bond energy is 1.67 eV in the lower ionosphere.

a. Identify the maximum wavelength a photon may have and still be capable of severing the bond. 1 photon carries full bond energy ϵ

$$\lambda_{\max} = \frac{hc}{E_{\text{bond}}} = \frac{1239.84 \text{ eV} \cdot \text{nm}}{1.67 \text{ eV}} = 742.4 \text{ nm}$$

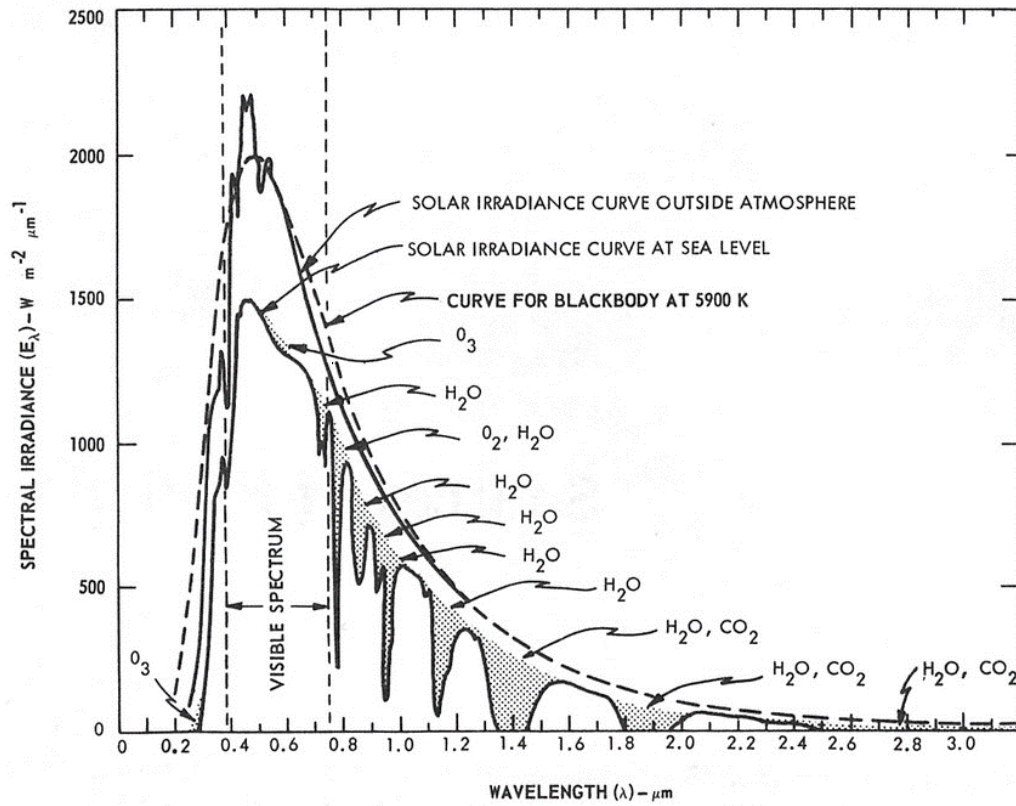
$$f_{\min} = \frac{c}{\lambda_{\max}} = \frac{2.998 \times 10^8}{742.4 \times 10^{-9}} = 4.038 \times 10^{14} \text{ Hz}$$

$$\lambda_{\max} = 742.4 \text{ nm} \quad (f_{\min} = 404 \text{ THz})$$

b. Is this a significant part of the solar spectrum (i.e., > 50%)? You do not need to quantify your answer.

742 nm is at the red edge of the visible, so everything that qualifies is the whole visible band plus all the UV.

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6. (10 pts) What is the approximate range of plasma frequencies (in MHz) associated with a 1000 km altitude orbit? Why are we interested in knowing that range of frequencies?

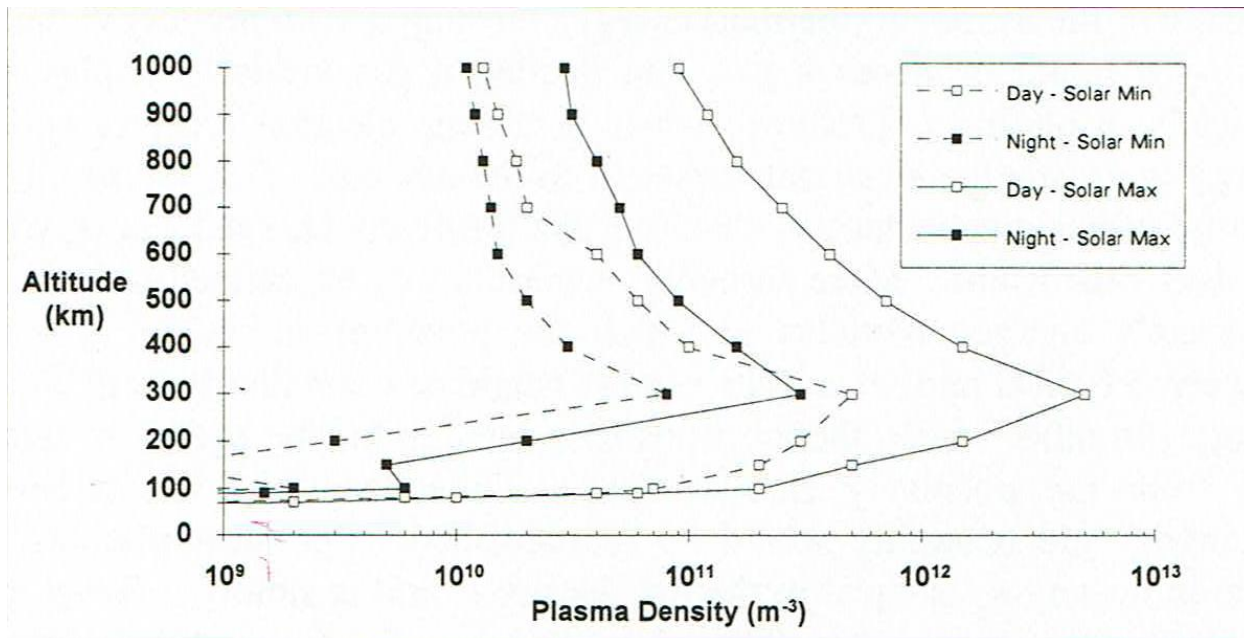
at 1000 km the range of plasma frequencies is 1-3 MHz

$$f_p = 8.98 \sqrt{n_e}$$

the range is from 10^{10} m^{-3} to 10^{11} m^{-3}

$$f_p = 8.98 \sqrt{10^{10}} = 8.98 \times 10^5 \text{ Hz} \approx 0.9 \text{ MHz}$$

$$f_p = 8.98 \sqrt{10^{11}} = 2.84 \times 10^6 \text{ Hz} \approx 2.8 \text{ MHz}$$



We care because it is a hard cutoff, not attenuation.
Below f_p the signal reflects and no link margin recovers it.

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7. (9 pts.) A CubeSat is planned for a 550 km sun-synchronous orbit and has a limited mass budget. The mission requires:

- 5-year lifetime
- Reliable communications
- Low development cost
- Commercial off-the-shelf electronics whenever possible

Select **three** environmental hazards that you would address first. Rank them from most to least important and justify your ranking.

For each hazard, recommend one mitigation strategy and discuss one tradeoff introduced by your solution.

Ranked by which hazard most certainly breaks a stated requirement over 5 years.

1. Radiation (TID + single-event effects on COTS). Polar sun-synchronous orbit crosses the auroral horns and SAA every rev. Dose only accumulates, and COTS parts have no rad-hard pedigree. This directly threatens reliable communications.

Mitigation: spot-shield the worst devices, latch-up current limiters, EDAC + watchdog.

Tradeoff: every reset is a comm outage; shielding eats the mass budget.

2. Neutral environment (atomic oxygen; drag is secondary). Continuous but not fatal. Drag from 550 km costs only ~40 km over 5 years and actually helps disposal. AO slowly erodes exposed polyimide.

Mitigation: AO-resistant coatings + low-drag attitude.

Tradeoff: coatings change α/ϵ ; low-drag attitude fights pointing.

3. Plasma (auroral charging/ESD). Dangerous but only episodic and cheapest to fix. Polar orbit crosses the auroral oval twice per rev; the arc, not the voltage, is the problem.

Mitigation: conductive exteriors bonded to a single ground.

Tradeoff: ITO costs power and money; bonding adds mass and labor.

MMOD is real at 550 km but a 3U has low collision probability and no mass left to shield it.

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8. (10 pts). What is the expected worst-case excess range and time delay for a satellite operating at 500 km in K-band. State your assumptions and show your work.

FREQUENCY BANDS AVAILABLE FOR SATELLITE COMMUNICATION

Band	Frequency Range	Total Bandwidth	General Application
L	1 to 2 GHz	1 GHz	Mobile satellite service (MSS)
S	2 to 4 GHz	2 GHz	MSS, NASA, deep space research
C	4 to 8 GHz	4 GHz	Fixed satellite service (FSS)
X	8 to 12.5 GHz	4.5 GHz	FSS military, terrestrial earth exploration, and meteorological satellites
Ku	12.5 to 18 GHz	5.5 GHz	FSS, broadcast satellite service (BSS)
K	18 to 26.5 GHz	8.5 GHz	BSS, FSS
Ka	26.5 to 40 GHz	13.5 GHz	FSS

$$\Delta R = \frac{40.31 \text{ TEC}}{f^2} = \frac{40.31 \times 10^{18}}{(1.8 \times 10^{10})^2} = 0.1244 \text{ m}$$

$$\Delta t = \frac{\Delta R}{c} = \frac{0.1244}{2.998 \times 10^8} = 4.15 \times 10^{-10} \text{ s}$$

Assumptions:
 $f = 18 \text{ GHz}$ (bottom of K-band), since excess range scales as $1/f^2$ and the lowest frequency is therefore the worst case.
 $\text{TEC} = 10^{18} \text{ electrons/m}^2$ (day, solar max, equatorial anomaly, disturbed). This is conservative because a 500 km satellite sits above the F2 peak, so part of the ionosphere is not in the path.
 Obliquity factor of 3 for a horizon-grazing pass.

$\Delta R = 12 \text{ cm}$
 $\Delta t = .42 \text{ ns Vertical}$
 $37 \text{ cm } \hat{=} 1.2 \text{ ns @ low elevation}$

Case	f	ΔR	Δt
Vertical band edge	18 GHz	12.44 cm	.415 ns
Vert top of band	26.5 GHz	5.74 cm	.191 ns
Slant (obliq 3)	18 GHz	37.3 cm	1.24 ns

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9. (15 pts.) A low-cost UCCS cube shaped satellite with each side 1 m^2 will operate in LEO at an altitude of 300 km. Assume an. ternal heat generation of 100 Watts and that only one side faces the sun and one side faces the Earth. Only black paint is used on all six sides. Black paint has an absorptivity of 0.975 and emissivity of 0.874. State any additional assumptions you make for this problem. There are no solar arrays on the satellite, so the satellite will use primary batteries that need to be kept between 0 and 15°C .

a. What are the equilibrium temperatures in the sun and in the eclipse?

$\sin \rho = \frac{6378}{6678} = 0.9551 \quad \sin^2 \rho = 0.9122$

$Q_{\text{solar}} = 0.975(1)(1367) = 1332.8 \text{ W}$
 $Q_{\text{oheto}} = 0.975(1)(0.9122)(465) = 413.6 \text{ W}$
 $Q_{\text{JA}} = 0.975(1)(0.9122)(237) = 210.8 \text{ W}$
 $\epsilon A_{\text{tot}} = 0.874(6) = 5.244 \text{ m}^2$

	Sun (W)	Eclipse (W)
Q_{solar}	1332.8	0
Q_{oheto}	413.6	0
Q_{JA}	210.8	210.8
Q_{internal}	100	100
Total	2057.2	310.8

$T_{\text{sun}} = \left(\frac{2057.2}{0.874(5.67 \times 10^{-8})(6)} \right)^{1/4} = 298.4 \text{ K} = 15.3^\circ\text{C}$
 $T_{\text{eclipse}} = 179.8 \text{ K} = -93.3^\circ\text{C}$

b. Will this be adequate for your mission? Explain.

No. Sunlit is .3 C past the 15 C ceiling with zero margin before end of life UV darkening.

c. If you could change up to four sides (the sides NOT facing the sun or Earth), what solution would you recommend? Use the lowest cost solution you can find. Clearly state your final equilibrium temperatures and the cost associated with your solution.

Hot case already uses the highest emissivity available (black paint). Cold case cannot be held passively, the two fixed black faces alone radiate too much, so heaters are mandatory.

The only way the same four faces can satisfy both is variable emissivity.

Recommendation: louvered radiators (emissivity 0.05 to 0.8) on the four sides + 304 W Kapton heaters.

$T_{\text{sun}} = 19.5^\circ\text{C}$ (louvers open)

$T_{\text{eclipse}} = 0^\circ\text{C}$ (louvers closed + heaters)

Mass 16.8 kg → cost \$420,000

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10. (10 pts.) **Discuss** three ways satellites can be designed to better survive MMOD impacts.

MMOD splits by size: track-and-dodge above ~10 cm, shield below ~1 cm, and design for the untrackable 1–10 cm band.

Whipple shields (sub-cm): thin bumper + standoff turns the particle into a debris cloud so the rear wall sees a spread load instead of a punch. Use only on tanks, batteries, and pressure vessels.

Configuration: smallest face into the velocity vector, critical hardware behind structure, bulkheads to limit damage. Nearly free but fights pointing and thermal layout.

Fault tolerance: separated redundancy, automatic isolation, and safe mode so a hit is survivable. Costs mass, power, and some availability.

Use all three: configuration first, shields on the critical items, fault tolerance for everything else.

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BONUS (+5 pts.)

11. Derive an expression for the rate of change of a satellite's altitude, i.e., $\dot{R}$, due to an applied ΔV . Assume the satellite's orbit is circular and use a method similar to the method we used to derive $\dot{R}$ due to drag.

$$1.) \text{ circular orbit } a=R: \mathcal{E} = -\frac{\mu m}{2R}$$

$$2.) \frac{d\mathcal{E}}{dt} = \frac{\mu m}{2R^2} \dot{R}$$

$$3.) \frac{d\mathcal{E}}{dt} = m \dot{v} \sqrt{\frac{\mu}{R}}$$

$$4.) \frac{\mu m}{2R^2} \dot{R} = m \dot{v} \sqrt{\frac{\mu}{R}} \Rightarrow \dot{R} = \frac{2R^2}{\mu} \dot{v} \sqrt{\frac{\mu}{R}}$$

$$\dot{R} = 2 \dot{v} \sqrt{\frac{R^3}{\mu}} = \frac{2 \dot{v}}{v} = \frac{2R \dot{v}}{v}$$